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Riparian Revegetation and Management Report R1

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1.0 Introduction

The purpose of this plan is to provide planting of riparian margins in accordance with Rule 7.14.2.5 and the standards in Appendix 7E of the Auckland Council District Plan – Operative Rodney Section 2011. The planting is to be located in the new rural residential lots of the proposed subdivision outlined in this application.

The total area to be protected is approximately 0.9ha, shown as Areas CO, CX, CY and CZ on the accompanying Scheme Plan.

2.0 Pre-Planting Assessment

2.1 Ecological District

The area to be revegetated is located in the Rodney Ecological District.

2.2 Characteristics of the soil

The soils are silty clay Waitemata Residual soils with generally good drainage. In Area CX and the east of Area CY, no water-logging of soils was noted around the stream channels. In the west of Area CY, the adjacent Area CZ, and CO, a wetland environment is present around the stream channels and species selection will need to reflect the higher moisture levels.

2.3 Topography and aspects of the area to be planted

The areas to be planted are generally flat or gently sloping around the stream channels.

2.4 Exposure of the site

Areas CZ and the west of Area CY are open and will be exposed to sun and wind damage, while the east of Area CY, CO and CX have some shading from existing trees within or nearby the areas to be planted. All areas will be exposed to frost over the winter months.

2.5 Extent of existing bush or native vegetation on the site and its species composition

Extensive areas of existing bush are present on the property, with the main vegetation types being either kauri, podocarp and broadleaf forest, or taraire forest on the lower slopes such as the southern parts of Area CU. Most of the existing native bush will be protected in association with this application. There is also a stand of native bush (Remiger Road Scenic Reserve) directly across the road from the planting areas, being in particular in close proximity to Area CX.

There are also some areas of regenerating kanuka forest on the property, being on the ungrazed edges of the existing mature bush. Common regenerating species include manuka, kanuka, mapou, mahoe, lancewood, and totara.

2.6 Vegetation in areas to be planted

In the Area CZ, CO, and western CY, the vegetation in the areas to be planted is largely exotic pasture species with some native and exotic rush (*Juncus*) species present. Some willows occur around the stream channel in the east of Area CY.

Area CX





3.0 Planting Plan Assessment

3.1 Site preparation for planting

Fencing

In accordance with Rule 7.14.2.5.2 of the plan, a stock proof fence shall be erected to protect the riparian margins. The standard for a permanent fence is a seven wire post and batten fence kept at least 10m back from the top of the watercourse banks.

Weed Control

The first step in a planting scheme is controlling the weeds present as it is much easier to eradicate weeds from a site before planting occurs than after planting has taken place. Initial and ongoing weed control as outlined in the Weed and Pest Management Report should be undertaken prior to planting, and ongoing monitoring carried out on a regular basis. Kikuyu occurs on the property, and ongoing control of this species will be particularly important to ensure this species does not smother the planted individuals, which may compromise the success of the planting.

Pest Control

Pest species which could potentially be a problem for planting in this area include rabbits, pukekos, and possums. Control methods for rabbits and pukeko are outlined below, and possums are covered in the Weed and Pest Management Report.

For rabbits and hares, removing possible nesting sites such as heaps of logs, blackberry and rank grass will reduce local numbers and reduce breeding success. Poisoning is the other option for which Pindone rabbit pellets are recommended. The placement of bait is vital to the success of the programme and signs of rabbit must be located. The most obvious is dropping piles or fresh digging. The bait should be applied within bait stations (e.g. Philproof), as these restrict unintended poisoning and keep the bait out of the weather so it lasts longer. Fence posts are convenient locations to attach these to. Turning over some earth bare near the bait station will act as an attractant for the rabbits. The bait should be left in place for at least three nights then ceased for several days. If there are still fresh rabbit signs after 14 days, repeat the process.

Pukeko can be a problem near wet areas because of their tendency to pull out young seedlings and tear apart newly planted individuals for food. Although they are a native species, they can be shot in season, or out of season with a DoC permit. Otherwise control should rely on keeping them away from the newly planted areas as much as possible. The most damage is likely to occur shortly after planting, so regular checks should be made for pukeko damage. If quickly discovered, the uprooted plants can generally be re-established successfully. If one or two pukeko are shot this will act as a deterrent to keep others away for a time, but they may return and are also active at night. Control will be an ongoing issue with the plants are establishing as a flock of pukeko can do a large amount of damage in a short space of time. Ongoing monitoring for pukeko damage along with any necessary control will be needed should they prove to be a significant problem.

3.2 Site planting

The overall goal of planting is to establish eco-sourced native vegetation that is tolerant of frost and drought and fast-growing, and will integrate in with existing vegetation on the site. These properties are possessed by several native pioneer species, which will form the bulk of the new planting. The recommended species are as follows:

- Manuka, kanuka, karamu, cabbage tree, flax, *Carex virgata*, totara, and kahikatea.

Note that when planting, a random pattern should be employed rather than planting in rows or the same distance apart all the time. A random pattern will result in a much more natural look to the planted area.

The planting areas are to be established at a density of one plant per $1m^2$ in accordance with plan guidelines for riparian areas, although existing vegetation in the north of Area CX ($800m^2$ approx.) will contribute to this density for that area.

Most plants will be of ¾ or 0.5 litre grade. In general, smaller plants will survive better over summer dry spells than larger grade plants. This is because of the larger root mass to leaf area volume than larger PB grade plants.

In the attached planting tables the spacing is the number of m^2/plant and then the average distance apart for that species. Following this results in the overall density shown. This provides a guide as to how far apart each species needs to be planted in order to form an even cover over the area to be planted. Therefore if just manuka is to be planted on one day, planting at the spacings shown will result in an even cover of this over the area. However, in practice some clustering may occur to give a more natural uneven appearance. Planting in lines or on a grid pattern should be avoided so a natural random look occurs.

Planting Methodology

Late autumn and winter are the best seasons for planting as most native plants are adapted to moist conditions and rainfall is higher than. When planting, one of the most important aspects for success is to ensure that root disturbance is kept to a minimum and to ensure that air spaces are not left between the root system and the soil. Therefore the seedlings need to be well heeled in. Planting also needs to occur into damp ground, ideally after rain in late autumn. A hole twice the size of the root ball should be dug and the seedling established in this. The roots should be loosened at the

base of the root mass prior to planting to aid root growth away from the root ball. Special care should be taken with manuka and kanuka as their roots are easily damaged. Some organic matter mixed with the soil at this time is helpful if the soil has a low organic matter content.

Fertilisation

With regard to fertilisation, for most areas a small amount of slow release fertilised added to the hole at the time of planting will increase plant performance. This fertiliser should be well mixed with the soil and not in direct contact with the plant roots, otherwise fertiliser burn can result. Alternatively, a fertiliser tablet placed in the hole on the upslope side of the plant is suitable. Dramatic difference in growth rates can occur between fertilised and unfertilised plants, and the extra costs will in all likelihood be more than offset by the reduced maintenance costs resulting from the plants rapidly reaching a height where releasing from grasses is no longer needed.

4.0 Management Plan

The table below outlines when activities such as initial planting, fertilisation, weed control and replacement planting should be undertaken. Planting should begin in late autumn once it has got wet or in winter. Ideally planting will be complete by July.

Annual maintenance cycle for establishing revegetation

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Year 1	*	* Ж		* # □	# Ж	#	#	Ж			*	
Year 2	*			∇ □ *	∇	∇	∇	Ж			*	
Year 3	*			∇	∇	∇	∇	Ж			*	
Year 4	*			∇				Ж			*	
Year 5	*			∇				Ж			*	

= Initial planting □ = Fertilisation if required

* = Weed control Ж = Pest control

∇ = Replacement planting of dead or poorly performing individuals

4.1 Planting Completion

By way of consent conditions, it will be a requirement to submit a report on planting completion after the planting is undertaken.

There will also be a requirement to monitor planting progress for a minimum of five years and until the survival rate is 90% or such time as 75% canopy closure across the entire planted area has been achieved. This will involve the attached monitoring sheets (see Appendix B) being on an annual basis, and submitted to Council.

5.0 Conclusion and Recommendations

Following the planting plan outlined in this report should result in the rapid and successful creation of native vegetated riparian margins. The planting and associated protection will result in a significant improvement over the existing situation so long as adequate maintenance is carried out while the plants are establishing.

This assessment is subject to the following recommendations:

- Stock are to be excluded from the areas to plant.
- The recommended planting and maintenance occurs.
- Any deceased plants are replaced.
- The recommended weed and pest control is undertaken and ongoing monitoring for weeds occurs to remove any future threats.
- A protective covenant over the area is registered on the titles to the land.

6.0 Limitations

This report has been prepared for the particular project described to us and its extent is limited to the scope of work agreed between the client and Cato Bolam Consultants Limited.

No responsibility is accepted by Cato Bolam Consultants Limited or its directors, servants, agents, staff or employees for the accuracy of information provided by third parties and/or the use of any part of this report in any other context or for any other purposes.

This report is for the use by the client only, and should not be used or relied upon by any other person or entity or for any other projects.

Appendix A: Plant Schedules

Plant Schedule for Riparian Planting Areas
(8,200m^2)

Botanical name	Common name	Grade	Spacing	Quantity
Leptospermum scoparium	Manuka*	¾ or 0.5 litre	(1/2.6m^2) 1.6m	3,200
Kunzea robusta	Kanuka	¾ or 0.5 litre	(1/3.7m^2) 1.9m	2,230
Coprosma robusta	Karamu	¾ or 0.5 litre	(1/8.2m^2) 2.9m	1,000
Cordyline australis	Cabbage tree*	¾ or 0.5 litre	(1/8.2m^2) 2.9m	1,000
Phormium tenax	Flax*^	¾ or 0.5 litre	(1/23.4m^2) 4.8m	350
Carex virgata	Purei*	¾ or 0.5 litre	(1/27.3m^2) 5.2m	300
Podocarpus totara	Totara	2L	(1/137m^2) 11.7m	60
Dacrycarpus dacroidoides	Kahikatea*	2L	(1/137m^2) 11.7m	60
Total				**8,200**

Average planting density 1/1m^2

* Plants suitable for water-logged soils

^ Should not be planted on the edge of the watercourse where flood risk present

Appendix B: Annual Monitoring Field Sheet for Native Planting

These sheets must be completed on an annual basis and submitted to Council for a five year period following the completion of the planting. The release of the various stages of the bond will require the satisfactory completion of these sheets and work as outlined in the maintenance schedule.

Administration

Date (dd/mm/yyyy): _____

Resource consent number: _____

Address: _____

Property owner and contact details: _____

_____

Has property changed owners in the last year, yes or no? _____

If yes, who was previous owner? _____

Survival Rate

Percentage survival: _____

Growth estimate (cm/year): _____

Percent ground cover (native vegetation): _____

Fertilisation

Date applied (dd/mm/yyyy): _____

What was used: _____

Areas applied: _____

_____

Quantity used: _____

Weed control

Date undertaken: _____

Sprays used: _____

Weeds targeted: _____

Areas targeted: _____

_____

Replacement planting

Date undertaken: _____

Species being replaced: _____

_____

Species planted: _____

_____

Number of plants replaced: _____

Problems

(for example are certain weeds proving difficult to control and detrimental to the planting, are animal pests causing significant problems)

Nature of problem: _____

_____

_____

_____

Possible solutions: _____

_____

_____

_____

Analysis of plant losses

(are losses greater than expected, are there any obvious reasons, are losses in certain areas, are certain species showing high losses, what are possible solutions?)

_____

_____

_____

_____